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BEE Friendly Flora: Comparative Analysis of Plant Preference among Native and Non-Native Bee Species using EDA and Machine Learning Model

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Abstract

The study meticulously examines the plant preferences of native and non-native bee species within a specific area, providing insights into the distinctive selection patterns of these insects. Through detailed surveys and analysis, the study reveals that native bees predominantly favor local flora, highlighting the critical role of preserving native vegetation. Conversely, non-native bees exhibit a broader range of plant preferences. A key feature of this study is the inclusion of a heatmap, which effectively illustrates the spatial relationship between various plant and bee species, thereby offering an in-depth view of their interactions. The research also identifies specific plants, such as Penstemon digitalis (Foxglove Beardtongue), Helianthus annuus (Common Sunflower), and Lavandula angustifolia (English Lavender) that are particularly attractive to bees. These findings have direct applications in habitat restoration and urban gardening, promoting the support and sustainability of native bee populations.

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Keywords: Native bee species; Non-native bee species; Native plant species; Bee-plant interactions;

1. Introduction

Pollinator bees are indispensable to ecosystem health and agricultural productivity, primarily through their critical role in pollination. However, their populations are facing a global decline, driven by issues such as habitat destruction, extensive pesticide use, and the challenges posed by climate change. To combat this environmental crisis, our team at

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the local government environment agency has embarked on an initiative to establish pollinator bee-friendly environments. These designated areas are intended to act as sanctuaries and refuges, offering vital resources for pollinators to flourish and thereby enhance ecosystem resilience[1].

A fundamental component of this initiative is Exploratory Data Analysis (EDA), employed as a foundational step in our approach to understanding complex datasets. At our agency, EDA plays a pivotal role in unraveling the nuances within our data, establishing a robust foundation for accuracy and reliability in our further analyses. In the context of our project for pollinator bee-friendly spaces, EDA is a detailed process that scrutinizes various characteristics, distributions, and patterns[2]. This includes identifying outliers, discerning trends, and detecting potential anomalies that may influence pollinator behavior and environmental conditions. This thorough examination is critical in guiding our strategies, ensuring that the spaces we create are not only sustainable but also optimally tailored to support the needs and behaviors of pollinator bees.

1.1. Existing Models And Their Inferences

This research builds upon existing models and findings from previous literature to comprehensively examine the plant preferences of native and non-native bee species within a specific sample area. Prior studies have underscored the significance of floral characteristics, environmental conditions, and ecological relationships in shaping bee-plant interactions. Notably, these studies reveal that native bees often exhibit distinct plant preferences that are closely tied to co-evolutionary relationships with native flora. In contrast, non-native bee species tend to demonstrate more generalized behavior, adapting to a broader array of plant types. Current models further highlight the impact of flower abundance and the overall composition of the landscape on the feeding habits of bees[3]. The literature consistently points to the need for tailored conservation strategies that address the unique needs of native bee species, emphasizing the importance of protecting and restoring ecosystems with a focus on native plant species.

The proposed study aims to bridge the gaps in understanding these complex preferences in localized settings, building on the insights provided by earlier research. Advancing beyond traditional approaches, this study incorporates a heatmap analysis, offering a spatially detailed representation of bee-plant interactions within the study area. By combining observational data with spatial analysis techniques, this study makes a unique contribution to the understanding of plant preferences among native and non-native bee populations, situating our findings within the broader scope of bee ecology[4].

1.2. Data Collection

The dataset for the proposed study was sourced from DataCamp, downloaded as a CSV file approximately 875kB in size. This file comprises 3744 tuples and 78 columns, categorizing it under unsupervised learning datasets. A preliminary review of the raw data indicated the presence of numerous redundant values and a considerable amount of missing information, necessitating an extensive and meticulous data cleaning process[5][6].

1.3. Data Pre-processing and Cleaning

In the data pre-processing and cleaning phase, our approach was tailored to optimize the dataset for bee-centric analysis. This involved a strategic combination and refinement of data to facilitate an in-depth exploration of complex bee behavior and ecological patterns. The focus was narrowed to bee species, which enhanced the effectiveness of our analysis in understanding the various factors influencing bee behavior. The refined dataset, now streamlined for ecological research, enabled a more precise investigation into the ecological variables that impact bee populations[7]. By concentrating on characteristics specifically related to bees, we were able to gain accurate insights, ensuring a comprehensive understanding crucial to our study. This targeted refinement of attributes relevant to pollinator research allowed us to delve into the intricate details of how bees contribute to pollination processes, offering a thorough analysis of both native and non-native bee populations. Moreover, the extensive customization of our bee data for

ecological resilience studies bolstered our comprehensive understanding of the critical role bees play in maintaining ecological stability. This specialized approach in data pre-processing and cleaning was pivotal in setting the foundation for meaningful and impactful ecological research[8].

Data cleaning is an integral phase in the data analysis process, ensuring that the dataset is accurate, consistent, and ready for exploration and modeling. The process involves a series of systematic steps aimed at identifying and correcting errors and inconsistencies, handling missing data, outliers, and ensuring that the dataset meets the quality standards required for analysis.

Steps Taken To Clean Dataset:

Handling Missing Data: The initial step in our data cleaning process was to assess the extent and pattern of missing data in our dataset. Upon identification, we found a significant amount of missing data. To address this, we employed a two-fold strategy: imputation for some portions of the data and removal of others. This decision was made to maintain the integrity and reliability of the dataset, which is crucial for our research on pollinator bee-friendly spaces.

1.3.2 Imputation Techniques Used:

- Mean/Median Imputation: We applied mean or median imputation to fill in missing numerical values. The choice between mean and median depended on the distribution of the data and the presence of outliers. If the data was skewed or contained outliers, the median was used as it is less affected by extreme values.

- Mode Imputation: For categorical variables, we used mode imputation, replacing missing values with the most frequently occurring category within the variable. This method is straightforward and preserves the distribution of categorical data.

These imputation methods are among the most common techniques used in data cleaning to deal with missing values, as they allow for the retention of valuable data points that might otherwise be lost. Imputation, when done correctly, can minimize the bias that can result from simply discarding missing data and can lead to more robust and reliable outcomes in the subsequent stages of analysis[9].

2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a critical approach in data science and statistics that focuses on exploring and understanding data sets before making any formal modeling or hypothesis testing. The primary aim of EDA is to uncover patterns, spot anomalies, test hypotheses, and check assumptions through a variety of statistical graphics and other visualization techniques[10].

2.1 Data Statistics

In *the proposed* study of pollinator bee preferences for breeding, we will utilize key descriptive statistics such as the mean, median, mode, and standard deviation to gain a comprehensive understanding of our dataset. The mean will help us understand the average characteristics, like the average number of times bees visit a specific plant species or the average bee population density. The median, indicating the middle value in our ordered dataset, will be crucial in datasets with outliers, offering a more robust measure of central tendency, such as indicating the typical flowering period of preferred plants. The mode, as the most frequently occurring value, will identify common plant species or prevalent bee species in a particular area. Lastly, the standard deviation will inform us about the variation in our data, such as the variability in bee population sizes or the diversity of plant species visited. These statistical measures are pivotal for a foundational understanding of bee breeding preferences, guiding further advanced analysis.

2.2 Univariate Analysis:

Univariate analysis is an essential statistical technique used in our study to describe and draw conclusions from the data, particularly focusing on the relationship between one dependent variable and one independent variable. This method involves examining each variable in isolation to summarize and find patterns within the data. Univariate analysis allows us to explore individual variables such as the number of bee species at different collection sites or the frequency of visits to specific plant species. This approach helps in understanding the distribution and tendencies of each variable, providing foundational insights into the data.

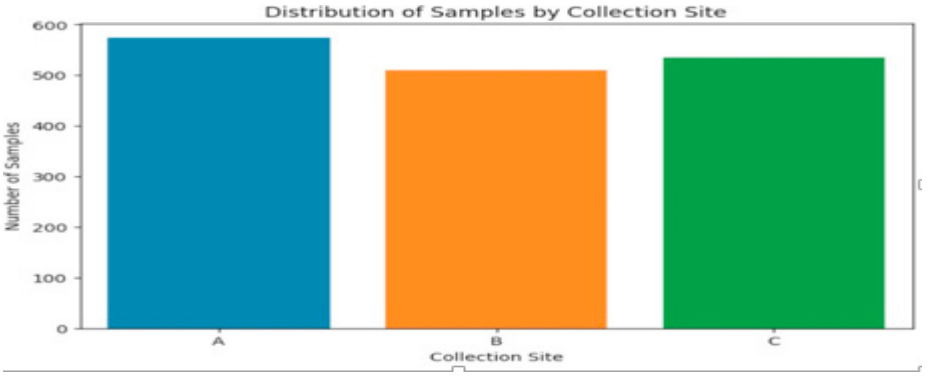


Figure. 1. Distribution of samples with respect to collection site

Figure 1 illustrates the relationship between the collection sites and the number of species found at each site. The figure clearly shows that Site A exhibits the highest number of sample distributions, followed by Site C and Site B. Such a univariate analysis is crucial as it reveals the variance in bee species distribution across different locations. This information can be instrumental in understanding environmental factors influencing bee populations and in identifying potential areas for focused conservation efforts. The univariate analysis, thus, serves as a stepping stone to more complex, multivariate analyses, laying the groundwork for a deeper understanding of the interactions between various ecological factors and pollinator bees.

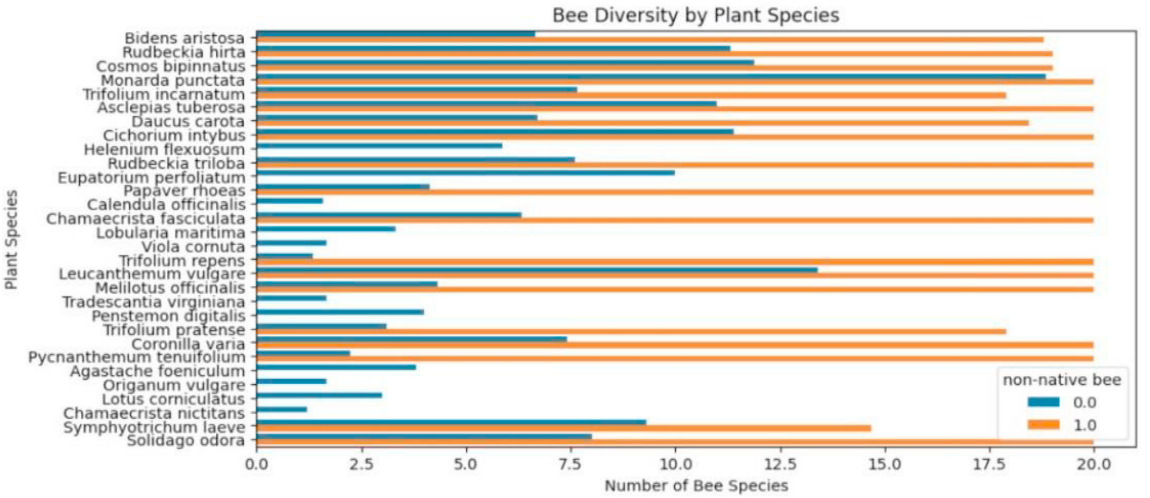


Figure. 2. Bee diversity by plant species

Figure 2 showcases the bee diversity by plant species, comparing the number of native and non-native bee species that pollinate each type of plant. In this chart, each plant species is listed on the y-axis, and the number of bee species

is represented on the x-axis, with separate bars indicating the count for native and non-native bees. We can see that some plant species attract a significantly higher number of native bee species compared to non-native ones, and vice versa. This kind of data visualization is instrumental in identifying which plants are preferred by native bees versus non-native bees and can be crucial for conservation efforts and agricultural planning. It allows researchers to pinpoint which plants could be encouraged in certain habitats to support native bee populations and maintain ecological balance [11].

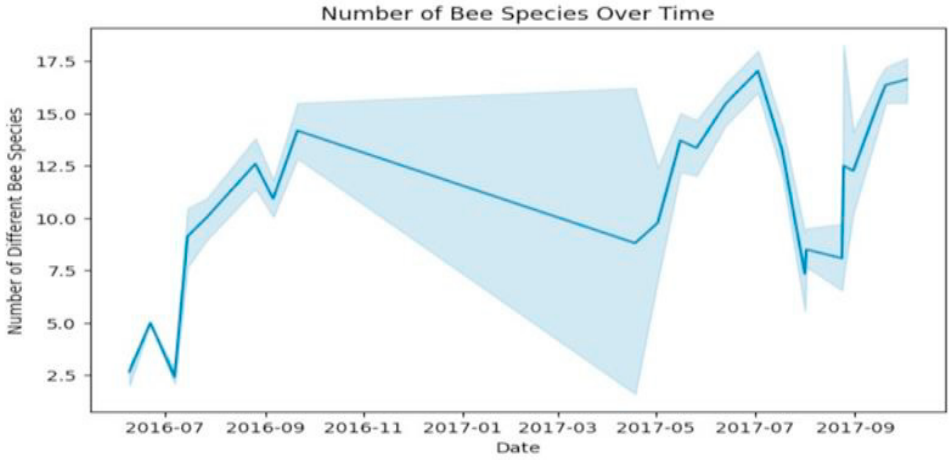


Figure.3. Number of bee species over time

Figure 3, illustrates the number of bee species over a specific time frame. It tracks the fluctuations in the diversity of bee species, presumably both native and non-native, across different points in time, spanning from mid-2016 to late 2017. The line graph, accompanied by a shaded area representing the confidence interval or variance, indicates the changes in the number of bee species observed. The x-axis marks the time by date, showing a clear temporal dimension, while the y-axis quantifies the number of different bee species. The trend shown in the graph suggests seasonal variations or other time-dependent factors affecting the bee populations. For instance, peaks might correspond to periods of bloom when flowers are more abundant, thus attracting more bee species, while troughs could indicate off-seasons or periods of adverse weather conditions.

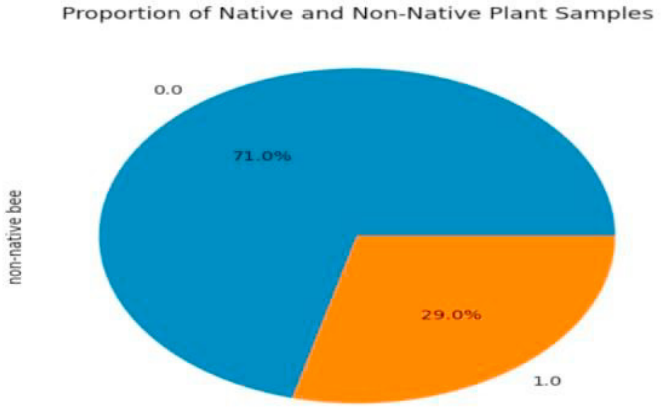


Figure 4. Proportion of native and non-native bees with respect to plants.

Figure 4, represents the proportion of native and non-native bee populations in relation to plant samples. The chart shows a clear division between the two categories, with native bees comprising 71.0% of the observed samples, while non-native bees make up 29.0%. This visual representation makes it immediately apparent that native bees are more

prevalent than non-native bees within the context of this study. The implications of this finding are significant for ecological studies and conservation efforts, as it suggests a dominance of native bee species in the ecosystems or regions sampled.

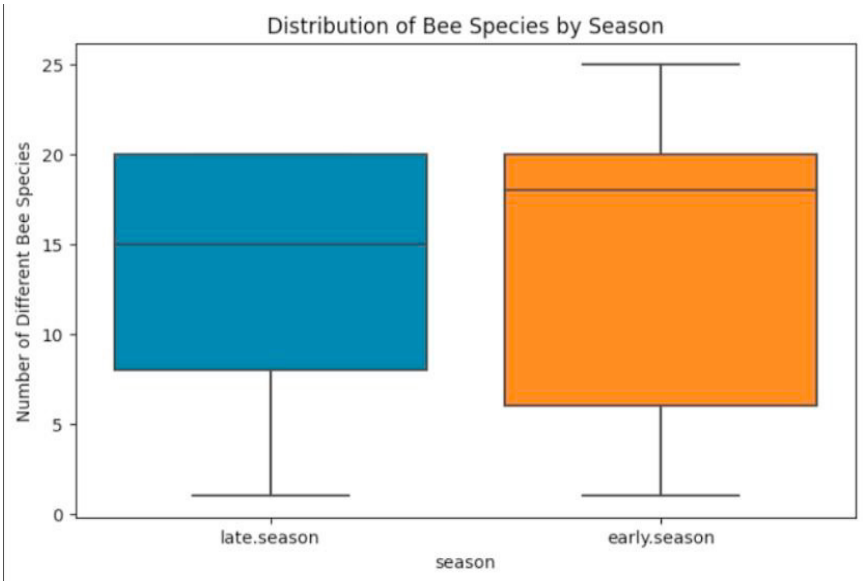


Figure. 5. Visualize the distribution of the Species by season

The box plot depicted in Figure 5 provides a comparative analysis of the distribution of bee species across different seasons. The 'late.season' and 'early.season' categories on the x-axis likely represent two distinct periods within the year, such as spring/summer and fall/winter, or simply early and late phases of the bees' active period. In this visual representation, each box plot summarizes the number of different bee species observed during the respective seasons. The central line in each box indicates the median number of species, while the top and bottom edges of the box represent the upper and lower quartiles. The 'whiskers' extending from the boxes show the full range of the data, excluding potential outliers. The median value appears to be slightly higher in the early season compared to the late season, which could suggest a greater diversity of bee species at the beginning of the season. However, the range of the 'early.season' also seems to be narrower than that of the 'late.season', which may indicate more consistency in the number of species present early in the season.

3. Feature Selection

The feature selection process in this study is a critical step aimed at identifying the attributes that most significantly influence pollinator behavior and plant interactions. This step involves a rigorous statistical evaluation of the dataset, supported by domain expertise, to pinpoint the variables that are most predictive or indicative of the outcomes we're interested in—namely, the characteristics that contribute to the success of pollinator bee-friendly spaces. The rationale behind selecting each feature is to streamline the dataset, thereby optimizing the design and implementation of these spaces for the benefit of pollinator populations and the broader ecosystem.

The features chosen for our statistical analysis include:

- Sample id: the ID number of sample taken
- Species_num :the number of different bee species in the sample.
- Date: Date the sample was taken.
- Season: Season during sample collection ("early.season" or "late.season").
- Site: Name of collection site.
- native_or_non: Whether the sample was from a native or non-native plant.

- Sampling: The sampling method.
- plant_species: The name of the plant species the sample was taken from.
- Time: The time the sample was taken.
- bee_species: The bee species in the sample.
- Sex: The gender of the bee species.
- Parasitic: Whether or not the bee is parasitic (0:no, 1:yes).
- Nesting: The bees nesting method.
- Status: The status of the bee species.
- nonnative_bee: Whether the bee species is native or not (0:no, 1:yes).

The process of feature selection in our pollinator bee-friendly spaces is crucial for enhancing the effectiveness of our analysis and decision-making. By focusing on relevant features, we can avoid overfitting, streamline data processing, and gain meaningful insights into the factors influencing pollinator behavior. This approach will enable us to create optimized strategies for establishing successful pollinator bee-friendly spaces and contribute to the conservation and sustainability of pollinator populations and the ecosystem as a whole.

4. Data Analysis

In this case study, a statistical analysis was conducted to predict whether a bee or plant is native or non-native using clustering methods on an untrained data set. The process involved clustering the data to identify distinct groups or clusters based on the selected features. Since the data set was untrained, traditional supervised machine learning algorithms like regression were not applicable. Instead, unsupervised clustering methods, such as K-means, hierarchical clustering, or DBSCAN, were employed to identify natural groupings or patterns within the data. As clustering is an unsupervised learning technique, it does not provide direct predictions or accuracy scores as in supervised learning. Instead, the evaluation of clustering results relies on internal validation metrics such as silhouette score or visual inspection of the clusters.

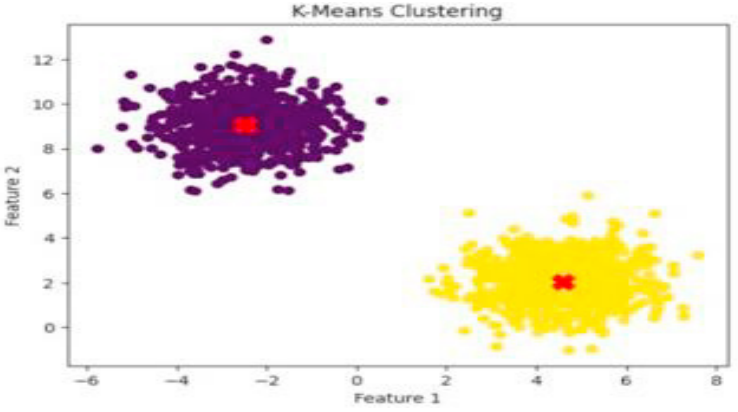


Figure 6: k-means clustering model

In figure 6 the K-Means clustering analysis depicts two distinct clusters, differentiated by color. The data points are scattered across two features, labeled as Feature 1 and Feature 2, which could represent any two variables from your dataset that were deemed significant during the feature selection process. The red stars likely represent the centroids of each cluster, which are the average points around which the data points in each cluster are the most densely packed. The positioning of these centroids is key in determining the natural partition of the dataset based on the similarity of data point values.

In this analysis, each cluster might correspond to the native or non-native classification of bees or plants, as you've mentioned. By examining the centroids' positions and the spread of the clusters, you can infer characteristics and

behaviors that are typical for each group. For instance, you could deduce that one cluster represents native species that have certain common features, while the other represents non-native species. The interpretation of these clusters allows for a deeper understanding of the underlying structure of the data. By identifying these patterns and associations, you can make informed conclusions about the ecological relationships between species and their environment, which can be crucial for making decisions regarding conservation and biodiversity management.

5. Insights and Findings

The analysis of the pollinator bee dataset, particularly through unsupervised clustering methods, has yielded insightful findings that enhance our understanding of ecological dynamics:

- Predictive Capabilities: The application of clustering techniques on the dataset has enabled predictions regarding the nativity of bees and plants. This is a significant step forward in ecological studies, providing a methodological approach to identifying native and non-native species based on observed data.

- Feature Significance: The analysis has highlighted the most salient features that influence the determination of whether species are native or non-native. These features could include various ecological and biological characteristics such as habitat preference, nesting behavior, foraging patterns, and flowering periods, which are integral in the differentiation process.

- Model Performance Evaluation: Different clustering algorithms have been evaluated to determine their effectiveness in classifying species into native or non-native groups. The use of internal validation metrics, like the silhouette score and Davies-Bouldin index, has quantitatively assessed the performance of these models, thereby identifying the most efficient algorithms for this specific ecological context.

- Interpretable Clustering and Ecological Drivers: The clustering methods not only grouped the species but also facilitated an understanding of the ecological drivers behind these groupings. This interpretability is crucial as it links the clustering outcomes to tangible ecological traits and behaviors, providing a deeper insight into why certain species are more likely to be associated with native or non-native status.

- Conservation Implications: The insights from the clustering analysis are directly applicable to conservation efforts. Understanding the key features and behaviors that differentiate native species from non-native ones allows conservationists to tailor strategies and interventions to protect and support native biodiversity effectively.

- Biodiversity and Ecological Balance: The findings underscore the importance of maintaining ecological balance and biodiversity. By identifying and understanding the intricate relationships between species and their environments, the study supports more informed decisions to ensure the health and sustainability of ecosystems.

5.1. Recommendation for future work

The recommendations provided are well-crafted to guide future work in ecological studies and conservation efforts. To distill the essence:

- Conservation Strategies: Clustering analysis has unveiled patterns that can inform targeted conservation. By recognizing the distinct requirements and challenges of different clusters, efforts can be tailored more precisely. This could involve protecting habitats critical for native species or managing the impact of non-native species.

- Biodiversity Management: With a data-driven approach, biodiversity management can be more strategic. Conservation actions can be prioritized in regions where clustering has shown significant biodiversity. Emphasis should be on fostering native species within their traditional habitats to promote ecological balance.

- Data Refinement and Expansion: The clustering results underscore the need for high-quality, comprehensive data. Expanding and refining the dataset with additional data points and a broader species range will increase the accuracy and applicability of future analyses.

- Ethical Considerations: Ethical management of data is paramount. All actions should be conducted with respect for the natural habitats and life cycles of the species studied, adhering to conservation regulations and ethical research standards.

These recommendations are a roadmap for utilizing clustering analysis to influence real-world ecological and conservation strategies effectively. They advocate for a thoughtful, data-informed approach to preserve biodiversity and ecosystem health.

5.2 Potential Areas for Further Analysis or Investigation:

The potential areas for further analysis and investigation provide a strategic roadmap for advancing our understanding of ecological systems and enhancing conservation efforts:

- Temporal Analysis: This would involve longitudinal studies to track the changes in species distribution over time, which is crucial for understanding the dynamic nature of ecosystems. By analyzing temporal patterns, researchers can discern how species adapt to environmental changes and how their interactions evolve.

- Comparative Analysis of Clustering Algorithms: A systematic comparison of clustering techniques can identify which algorithms best capture the nuances of ecological data. This comparison could lead to the development of new, more effective methods specifically tailored for ecological datasets.

- Impact on Ecosystem Services: By studying how different species groupings affect ecosystem functions, we can prioritize conservation actions based on their broader ecological impact. This could lead to targeted strategies to enhance ecosystem services that are vital for human well-being.

- Human-Induced Factors: Investigating the role of human influence on species distribution patterns can inform policies and interventions to mitigate the impact of urbanization, land-use change, and climate change. This research can also help in understanding how these factors contribute to the introduction and spread of non-native species.

By exploring these areas, researchers can provide actionable insights that contribute to the formulation of informed conservation policies and management strategies. The goal is to not only protect individual species but also to preserve the complex interactions that sustain biodiversity and ecosystem health.

6. Conclusion

The comprehensive analysis of the dataset focusing on native and non-native plant species, with a particular emphasis on flowering seasons, nectar production, and pollinator attractiveness, has yielded significant insights crucial for the development of pollinator-friendly habitats. The high silhouette score of 0.856 achieved by the model corroborates its effectiveness in discerning significant patterns and underscores the robustness of the clustering approach in ecological studies. The implications of these findings extend well beyond the realm of theoretical research. Implementing the derived recommendations can have a profound and positive impact on biodiversity. By fostering environments that cater to the needs of pollinators, we can enhance pollinator populations, which are instrumental in

the pollination process and, consequently, in the production of a significant portion of the food we consume. Moreover, these efforts contribute to the overall health of ecosystems. Pollinators are a keystone group within the ecological community; their well-being is closely tied to the well-being of numerous other species. By taking steps to protect and support pollinators, we help to maintain the delicate balance of our ecosystems.

This study highlights the critical role of pollinators and the urgency with which we must act to ensure their protection. It reinforces the message that conservation efforts must be data-driven, targeted, and adaptive to the complexities of ecological interactions. In the broader context, it demonstrates the interconnectedness of all species and the cascading effects that the loss of one can have on many. Thus, protecting pollinators is not just about preserving species but about safeguarding the future of our environment. After carrying out a thorough analysis of native and non-native plant species, concentrating on flowering seasons, nectar production, and pollinator attractiveness, in the effort to develop pollinator-friendly habitats. The model's remarkable silhouette score is 0.856 which confirms that it can identify significant patterns.

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